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To cite this article: Nitis Mukhopadhyay (2022) Pairwise Independence May Not Imply Independence: New Illustrations and a Generalization, The American Statistician, 76:2, 184-187, DOI: [10.1080/00031305.2022.2039763](https://doi.org/10.1080/00031305.2022.2039763)

To link to this article: <https://doi.org/10.1080/00031305.2022.2039763>



Published online: 23 Mar 2022.



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Pairwise Independence May Not Imply Independence: New Illustrations and a Generalization

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ABSTRACT

A number of standard textbooks that are followed in a junior/senior level course or in a first-year graduate level course in mathematical statistics and probability, routinely include one single basic illustration, obviously in its variant forms, to highlight an important point: *pairwise independence* may not imply (*mutual independence*). We earnestly believe that beginning students appreciate more examples to clarify these key issues. Hence, we hope that our new sets of nontrivial illustrations from [Section 2](#) will help our audience. Next, in [Section 3](#), we extend the notion to *q-wise independence* with a large set of illustrations using both discrete and continuous random variables showing that *q-wise independence* may not imply (*mutual independence*). We reasonably assure that this discourse is immediately accessible to juniors/seniors and first-year graduate students.

ARTICLE HISTORY

Received May 2021
Accepted February 2022

KEYWORDS

Continuous distributions;
Discrete distributions;
Mutual independence;
Pairwise independence;
q-Wise independence

1. Introduction

Relationships between pairwise independence and mutual independence may quickly become unsettling for readers to clearly understand the topic in junior/senior and first-year graduate courses in mathematical statistics and probability. In these elementary courses, a quick illustration is normally given to highlight that pairwise independence may not imply mutual independence.

One considers tossing a fair coin two times and defines random variables:

$$\begin{aligned} X_1 &= \begin{cases} 1 & \text{if } A_1 \text{ occurs (we observe a head on the first toss)} \\ 0 & \text{otherwise,} \end{cases} \\ X_2 &= \begin{cases} 1 & \text{if } A_2 \text{ occurs (we observe a head on the second toss)} \\ 0 & \text{otherwise,} \end{cases} \\ X_3 &= \begin{cases} 1 & \text{if } A_3 \text{ occurs (we observe same outcome on} \\ & \text{both tosses)} \\ 0 & \text{otherwise.} \end{cases} \end{aligned} \quad (1.1)$$

Clearly, A_i, A_j are independent events for *every* pair of indices i and j , $1 \leq i \neq j \leq 3$, but the three events A_1, A_2, A_3 are not mutually independent. This is restated as an obvious conclusion that the random variables X_1, X_2, X_3 are pairwise independent, but they are not mutually (or jointly) independent.

Many classic texts including those of Ross (1997, p. 11), Casella and Berger (2002, pp. 26–27), Zacks (2014, pp. 57–58), and Hogg et al. (2019, pp. 130–131) discuss variants of this example. Simple but nontrivial variations of (1.1) and ensuing illustrations are rather very few and far between to nearly nonexistent as far as we can tell. A clear exception is Example 3.5.4 in Mukhopadhyay (2000, pp. 126–127). We feel strongly that a large collection of easily available illustrations is essential for readers to clearly grasp the topic.

[Section 2](#) addresses that void by constructing a breadth of new and substantial set of illustrations. [Section 3](#) introduces a concept called *q-wise independence* when we have $q + 1$ random variables ($q \geq 2$) under consideration. Again, we provide a different and new kind of construction to exhibit a large class of examples to justify that *q-wise independence* may not imply mutual independence with a large set of illustrations using both discrete and continuous random variables. We hope that this presentation will directly benefit readers especially since our illustrations from [Sections 2](#) and [3](#) also create ample opportunities for them to explore and improvise further, both inside and outside of classrooms.

2. Pairwise Independence and Mutual Independence

We say that the random variables X_1, X_2, X_3 are *pairwise independent* if and only if X_i, X_j are independently distributed for

every pair of indices i and j , $1 \leq i \neq j \leq 3$. We also say that X_1, X_2, X_3 are (mutually) independent if and only if the following holds:

$$f_{X_1, X_2, X_3}(x_1, x_2, x_3) = \prod_{i=1}^3 f_{X_i}(x_i), \text{ for all data points } (x_1, x_2, x_3). \quad (2.1)$$

The customary example from (1.1) shows that *pairwise independence* may hold, but mutual independence may not. Sections 2.1 and 2.2 show completely new kinds of constructs where *pairwise independence* holds, but mutual independence does not. A curious learner will be able to explore numerous other possibilities. We include some directions in Section 2.2.

2.1. Illustration 1

We fix three arbitrary, positive, integrable functions $a_i(x)$, different from one another with probability 1 (w.p.1), each defined on the support $(0, c)$ with $c > 0$. We also assume that $\int_0^c a_i(x) dx = 1$, $i = 1, 2, 3$, that is, each $a_i(x)$ happens to be a *probability density function* (pdf) on $(0, c)$. We will keep $a_i(x)$'s rather simple in their appearance, requiring very little to nearly no effort to integrate them.

Let $\mathbf{x} = (x_1, x_2, x_3)$ and create a function of three variables as follows:

$$f^{[3]}(\mathbf{x}; c) \equiv f^{[3]}(\mathbf{x}) = \frac{1}{4} \left[a_1(x_1) a_2(x_2) a_3(x_3) + c^{-2} \left\{ a_1(x_1) + a_2(x_2) + a_3(x_3) \right\} \right], \quad (2.2)$$

so that $f^{[3]}(\mathbf{x}; c) > 0$ for all $\mathbf{x} \in (0, c)^3$. The superscript used in $f^{[3]}(\mathbf{x}, c)$ indicates that we have three random variables in the full set under consideration. Since c is held fixed, we often drop c from the label.

From (2.2), we easily obtain the total probability:

$$\int_0^c \int_0^c \int_0^c f^{[3]}(\mathbf{x}) dx_1 dx_2 dx_3 = \frac{1}{4} [1 + c^{-2} \{3c^2\}] = 1. \quad (2.3)$$

Now, we may conceptualize random variables X_1, X_2, X_3 whose joint pdf is given by $f^{[3]}(\mathbf{x})$ on $(0, c)^3$.

The associated one-dimensional marginals are given by:

$$\begin{aligned} f_1^{[3]}(x_1) &= \int_0^c \int_0^c f^{[3]}(\mathbf{x}) dx_2 dx_3 \\ &= \frac{1}{4} [a_1(x_1) + c^{-2} \{c^2 a_1(x_1) + 2c\}] \\ &= \frac{1}{2} [c^{-1} + a_1(x_1)], \text{ and similarly,} \\ f_2^{[3]}(x_2) &= \frac{1}{2} [c^{-1} + a_2(x_2)], \\ f_3^{[3]}(x_3) &= \frac{1}{2} [c^{-1} + a_3(x_3)], \end{aligned} \quad (2.4)$$

for all $\mathbf{x} \in (0, c)^3$.

Next, we find the two-dimensional marginals as follows:

$$\begin{aligned} f_{ij}^{[3]}(x_i, x_j) &= \int_0^1 f^{[3]}(\mathbf{x}) dx_k = \frac{1}{4} \{c^{-1} + a_i(x_i)\} \{c^{-1} + a_j(x_j)\} \\ &= f_i^{[3]}(x_i) f_j^{[3]}(x_j), \end{aligned} \quad (2.5)$$

with $i \neq j \neq k = 1, 2, 3$ and for all $\mathbf{x} \in (0, c)^3$.

Combining (2.2)–(2.5) and using the notion of pairwise independence, we conclude that X_i, X_j are independently distributed for every pair of indices i and j , $1 \leq i \neq j \leq 3$, but the full collection $\{X_1, X_2, X_3\}$ is easily shown to be *dependent*. This construct is completely different from what we initially had in (1.1).

2.2. Illustration 2

Again, we begin with three arbitrary, positive, integrable functions $a_i(x)$, different from one another w.p.1, each defined on $(0, 1)$, so that $\int_0^1 a_i(x) dx = 1$, $i = 1, 2, 3$. Let us create a function of three variables as follows:

$$\begin{aligned} g^{[3]}(\mathbf{x}) &= \frac{1}{32} \left[5a_1(x_1) a_2(x_2) a_3(x_3) + 3a_1(x_1) a_2(x_2) \right. \\ &\quad \left. + 3a_1(x_1) a_3(x_3) + 3a_2(x_2) a_3(x_3) \right. \\ &\quad \left. + 5a_1(x_1) + 5a_2(x_2) + 5a_3(x_3) + 3 \right], \end{aligned} \quad (2.6)$$

with $\mathbf{x} \in (0, 1)^3$. This is a joint pdf on the support $(0, 1)^3$ and we conceptualize random variables X_1, X_2, X_3 whose joint pdf is $g^{[3]}(\mathbf{x})$.

With the help of simple integration techniques like those used in Section 2.1, we easily verify that the associated one- and two-dimensional marginals are given as follows:

$$\begin{aligned} g_i^{[3]}(x_i) &= \frac{1}{2} [1 + a_i(x_i)] \quad \text{and} \\ g_{ij}^{[3]}(x_i, x_j) &= \frac{1}{4} [1 + a_i(x_i)] [1 + a_j(x_j)] \\ &= g_i^{[3]}(x_i) g_j^{[3]}(x_j), \text{ for all } i \neq j = 1, 2, 3 \\ &\quad \text{and for all } \mathbf{x} \in (0, 1)^3. \end{aligned} \quad (2.7)$$

Combining (2.6)–(2.7) and using the notion of pairwise independence, we conclude that X_i, X_j are independently distributed for every pair of indices i and j , $1 \leq i \neq j \leq 3$, but the full collection $\{X_1, X_2, X_3\}$ is easily shown to be *dependent*.

These constructions will let a reader incorporate numerous choices of $a_i(x_i)$ functions. Here is another interesting construction: On the support $(0, 1)^3$, we may revisit $f^{[3]}(\mathbf{x}; c = 1)$ from (2.2) and $g^{[3]}(\mathbf{x})$ from (2.6), thereby defining an alternative joint pdf constructed as follows:

$$h^{[3]}(\mathbf{x}) = \alpha f^{[3]}(\mathbf{x}; c = 1) + (1 - \alpha) g^{[3]}(\mathbf{x}) \text{ with } 0 < \alpha < 1. \quad (2.8)$$

This is a mixture distribution. Again, one can easily verify under (2.8) that X_i, X_j are independently distributed for every i and j , $1 \leq i \neq j \leq 3$, but X_1, X_2, X_3 are *dependent*. That said, a substantially more general construction is introduced in Section 3.

3. A Generalization: q -Wise Independence and Mutual Independence

Let us suppose that we have a collection of random variables $\{X_1, \dots, X_{q+1}\}$ with $q \geq 2$ and pursue a notion which we refer to as q -wise independence defined as follows.

Definition 3.1. $X_1, \dots, X_{q+1}$ are called q -wise independent if and only if every q -tuple $X_{i_1}, \dots, X_{i_q}$ consists of independently distributed random variables whatever be the q indices $i_1, \dots, i_q, 1 \leq i_1 \neq \dots \neq i_q \leq q+1$.

Section 2.1 dealt with the case when $q = 2$, that is pairwise independence. It is immediate to conclude that if $X_1, \dots, X_{q+1}$ are q -wise independent, then $X_1, \dots, X_{p+1}$ are also p -wise independent, $2 \leq p < q$.

3.1. q -Wise Independence May Not Imply Mutual Independence: Continuous Distributions

We fix $q+1$ positive, symmetric, and integrable functions $b_i(x)$, all different from one another w.p.1, for $x \in (-c, c)$ such that

$$\int_{-c}^c b_i(x) dx = 1, i = 1, \dots, q+1, c > 0.$$

In other words, each $b_i(x)$ is a symmetric pdf sitting on the support $(-c, c)$. For simplicity, we suggest considering:

$$b_i(x) = \begin{cases} \frac{1}{2}(1 + 2\delta_i)c^{-(2\delta_i+1)}x^{2\delta_i} & \text{if } -c < x < c \\ 0 & \text{otherwise,} \end{cases} \quad (3.1)$$

where δ_i 's are distinct positive integers, that is $b_i(x)$ is clearly an even function, $i = 1, \dots, q+1$.

Now, we construct:

$$f^{[q+1]}(\mathbf{x}) = \begin{cases} [1 + c^{-(q+1)} \prod_{j=1}^{q+1} x_j] \prod_{i=1}^{q+1} b_i(x_i) & \text{if } -c < x_1, \dots, x_{q+1} < c \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

We note that the term $1 + c^{-(q+1)} \prod_{j=1}^{q+1} x_j > 0$ for all $-c < x_1, \dots, x_{q+1} < c$, that is, $f^{[q+1]}(\mathbf{x})$ from (3.2) is always nonnegative. A reader can show that $f^{[q+1]}(\mathbf{x})$ integrates to 1 on the product space $(-c, c)^{q+1}$ which implies that $f^{[q+1]}(\mathbf{x})$ is a joint pdf.

Next, let us borrow the customary notation from regression analysis when one random variable is removed or deleted at-a-time from $\mathbf{X}$, namely,

$$\mathbf{X}_{(-1)} = (X_2, \dots, X_{q+1}),$$

$$\mathbf{X}_{(-i)} = (X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_{q+1}) \quad \text{for } i = 2, \dots, q,$$

$$\mathbf{X}_{(-q+1)} = (X_1, \dots, X_q),$$

Then, a reader can show that (a) the marginal pdf of X_i is $b_i(x_i)$, (b) the joint pdf of $\mathbf{X}_{(-i)}$ is $\prod_{j=1, j \neq i}^{q+1} b_j(x_j)$ for every fixed $i, i = 1, \dots, q+1$.

Combining these ideas, we can claim that $\mathbf{X}_{(-i)}$ consists of q independent random variables for every fixed $i, i = 1, \dots, q+1$, that is $X_1, \dots, X_{q+1}$ are q -wise independent, but the full collection of $q+1$ random variables $X_1, \dots, X_{q+1}$ are dependent.

Remark 3.1. One can check that the multiplier $[1 + c^{-(q+1)} \prod_{j=1}^{q+1} x_j]$ used in defining $f^{[q+1]}(\mathbf{x})$ from (3.2) may be replaced by a term such as (i) $1 + \rho c^{-\sum_{i=1}^{q+1} d_i} \prod_{i=1}^k x_i^{d_i}$ or (ii) $1 + \rho \prod_{i=1}^k \sin(x_i^{d_i})$,

with $\rho \in (-1, 1) - \{0\}$ and d_i 's positive odd integers. Then, $X_1, \dots, X_{q+1}$ are q -wise independent, but the full collection of $q+1$ random variables $X_1, \dots, X_{q+1}$ are dependent.

3.2. q -Wise Independence May Not Imply Mutual Independence: Discrete Distributions

We fix $q+1$ positive, symmetric, and probability mass functions (pmf) $b_i(x)$, all different from one another w.p.1, for $x \in \{0, \pm 1, \pm 2, \dots, \pm k\}$ where k is a fixed positive integer. Let us begin with three sets of possible examples of such functions defined as follows:

(i) $b_i(x)$

$$= \begin{cases} (1 - \theta_i)^k & \text{if } x = 0 \\ \frac{1}{2} \binom{k}{|x|} \theta_i^{|x|} (1 - \theta_i)^{k-|x|} & x = \pm 1, \dots, \pm k; \end{cases}$$

(ii) $b_i(x)$

$$= \begin{cases} \theta_i \left\{ 1 - (1 - \theta_i)^{k+1} \right\}^{-1} & \text{if } x = 0 \\ \frac{1}{2} \theta_i (1 - \theta_i)^{|x|} \left\{ 1 - (1 - \theta_i)^{k+1} \right\}^{-1} & x = \pm 1, \dots, \pm k; \end{cases}$$

(iii) $b_i(x)$

$$= \begin{cases} e^{-\theta_i} \left\{ 1 - e^{-\theta_i} \sum_{j=k+1}^{\infty} \theta_i^{j+1} ((j+1)!)^{-1} \right\}^{-1} & \text{if } x = 0 \\ \frac{1}{2} e^{-\theta_i} \theta_i^{|x|} (|x|!)^{-1} \left\{ 1 - e^{-\theta_i} \sum_{j=k+1}^{\infty} \theta_i^{j+1} (j!)^{-1} \right\}^{-1} & x = \pm 1, \dots, \pm k; \end{cases} \quad (3.3)$$

where $\theta_1, \dots, \theta_k$ are different from one another with (a) $0 < \theta_i < 1$ in parts (i) and (ii), (b) $0 < \theta_i < \infty$ in part (iii), $i = 1, \dots, k$. The expressions from (3.3) parts (i)–(iii) will clearly remind a binomial, geometric and Poisson distribution, respectively.

Now, in the spirit of (3.2), we may construct the following nonnegative function:

$$f^{[q+1]}(\mathbf{x}) = \begin{cases} [1 + k^{-(q+1)} \prod_{j=1}^{q+1} x_j] \prod_{i=1}^{q+1} b_i(x_i) & \text{if } \mathbf{x} \in \mathcal{A} = \{0, \pm 1, \pm 2, \dots, \pm k\}^{q+1} \\ 0 & \text{otherwise.} \end{cases} \quad (3.4)$$

with $b_i(x_i)$'s coming from (3.3). A reader can show that $\sum_{\mathbf{x} \in \mathcal{A}} f^{[q+1]}(\mathbf{x}) = 1$ which would imply that $f^{[q+1]}(\mathbf{x})$ is a joint pmf on the set $\mathcal{A}$.

Then, a reader can show that (a) the marginal pmf of X_i is $b_i(x_i)$, and (b) the joint pmf of $\mathbf{X}_{(-i)}$ is $\prod_{j=1, j \neq i}^{q+1} b_j(x_j)$ for every fixed $i, i = 1, \dots, q+1$. Combining these ideas, we can claim that $X_1, \dots, X_{q+1}$ are q -wise independent discrete random variables, but the full collection $X_1, \dots, X_{q+1}$ are dependent.

Readers are encouraged to explore the phenomenon “ q -wise independence may fail to imply mutual independence” with numerous concrete examples led by the construction given by (3.2) and (3.4) combined with numerous other possibilities opened in view of Remark 3.1. We earnestly hope that these may help readers to grasp this new concept called q -wise independence. By fixing $q = 2$, one will readily construct more illustrations to supplement those suggested in Sections 2 and 3 showing that pairwise independence may not imply mutual independence.

Remark 3.2. A reviewer brought up a related issue of possible dependence along with a zero variance-covariance matrix, that is, with zero Pearson correlation coefficients. Our examples illustrate that, in general, a diagonal variance-covariance matrix need not imply that the random variables are independent. In these examples, one will note that other correlation coefficients, such as Spearman’s and Kendall’s are also zero. De Paula (2008) and Mukhopadhyay (2009) may be helpful.

Acknowledgments

Constructive comments generously shared by a reviewer, an Associate Editor, and the Editor helped in preparing this revised version. They are duly thanked. On Friday September 24, 2021, I had an opportunity to deliver an invited online presentation on this topic for the outreach program of the Calcutta Statistical Association which was attended by a large number of colleagues including many younger professionals from India and abroad. I thank Professor Dhiman Dutta, Professor Sumanta Adhya and their team for the arrangements. Professor Bikas K. Sinha and Professor

Debanjan Bhattacharjee wrote back encouraging thoughts. I decided to add Section 3.2 in response to a question raised by Professor Bhattacharjee. I express my personal gratitude to all.

Disclosure Statement

The author declares no potential conflict of interest with respect to research, authorship, and publication of this article.

Funding

The author received no financial support for this research, authorship, and publication of this article.

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